

AI-Powered Research Assistant Proposal

Project Objective: Build a Retrieval Augmented Generation (RAG) application capable of answering questions from a collection of research papers. Architecture: PDF Loader -> Text Splitter -> Embedding Model -> Vector Database -> Retriever -> LLM Tech Stack: LangChain, LangGraph, ChromaDB, OpenAI/Ollama Models, FastAPI. Deliverables: 1. Document Ingestion 2. Semantic Search 3. Conversational Interface 4. Source Citation Expected Outcome: A scalable AI assistant capable of retrieving and summarizing information from large document collections.